

Сетевой клиент

1.0

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Глава 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Глава 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

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Глава 3

Class Documentation

3.1 AuthManager Class Reference

Статический класс для криптографических операций

```
#include <auth_manager.h>
```

Static Public Member Functions

- static std::string [generateSalt](#) ()
- static std::string [computeHash](#) (const std::string &salt, const std::string &password)

Private Member Functions

- [AuthManager](#) ()=delete
- [~AuthManager](#) ()=delete

3.1.1 Detailed Description

Статический класс для криптографических операций

Definition at line 15 of file auth_manager.h.

3.1.2 Constructor & Destructor Documentation

3.1.2.1 AuthManager()

AuthManager::AuthManager () [private], [delete]

3.1.2.2 ~AuthManager()

AuthManager::~AuthManager () [private], [delete]

3.1.3 Member Function Documentation

3.1.3.1 computeHash()

```
std::string AuthManager::computeHash (
    const std::string & salt,
    const std::string & password ) [static]
```

Definition at line 26 of file auth_manager.cpp.

Referenced by ConnectionManager::authenticate().

3.1.3.2 generateSalt()

```
std::string AuthManager::generateSalt ( ) [static]
```

Definition at line 16 of file auth_manager.cpp.

Referenced by ConnectionManager::authenticate().

The documentation for this class was generated from the following files:

- src/[auth_manager.h](#)
- src/[auth_manager.cpp](#)

3.2 ClientParams Struct Reference

Структура для хранения параметров клиента

```
#include <user_interface.h>
```

Public Attributes

- std::string [server_address](#)
Адрес сервера
- uint16_t [port](#) = 33333
Порт сервера
- std::string [input_file](#)
Входной файл
- std::string [output_file](#)
Выходной файл
- std::string [config_file](#) = "~/config/vclient.conf"
Файл конфигурации

3.2.1 Detailed Description

Структура для хранения параметров клиента

Definition at line 14 of file user_interface.h.

3.2.2 Member Data Documentation

3.2.2.1 config_file

```
std::string ClientParams::config_file = "~/config/vclient.conf"
```

Файл конфигурации

Definition at line 19 of file user_interface.h.

Referenced by `UserInterface::UserInterface()`.

3.2.2.2 input_file

```
std::string ClientParams::input_file
```

Входной файл

Definition at line 17 of file user_interface.h.

Referenced by `UserInterface::UserInterface()`.

3.2.2.3 output_file

```
std::string ClientParams::output_file
```

Выходной файл

Definition at line 18 of file user_interface.h.

Referenced by `UserInterface::UserInterface()`.

3.2.2.4 port

```
uint16_t ClientParams::port = 33333
```

Порт сервера

Definition at line 16 of file `user_interface.h`.

Referenced by `UserInterface::UserInterface()`.

3.2.2.5 server_address

```
std::string ClientParams::server_address
```

Адрес сервера

Definition at line 15 of file `user_interface.h`.

Referenced by `main()`, and `UserInterface::UserInterface()`.

The documentation for this struct was generated from the following file:

- [src/user_interface.h](#)

3.3 ConnectionManager Class Reference

Класс для управления TCP-соединением с сервером

```
#include <connection_manager.h>
```

Public Member Functions

- [ConnectionManager](#) (const std::string &server_addr, uint16_t port)
- virtual [~ConnectionManager](#) ()
- void [authenticate](#) (const std::string &login, const std::string &password)
- std::vector< int32_t > [processVectors](#) (const std::vector< std::vector< int32_t >> &vectors)

Protected Member Functions

- virtual void [sendData](#) (const void *data, size_t size)
- virtual void [receiveData](#) (void *data, size_t size)
- virtual void [sendString](#) (const std::string &str)
- virtual std::string [receiveString](#) (size_t max_length=1024)

Protected Attributes

- int [socket_](#)
- std::string [server_addr_](#)
- uint16_t [port_](#)

3.3.1 Detailed Description

Класс для управления TCP-соединением с сервером

Definition at line 21 of file connection_manager.h.

3.3.2 Constructor & Destructor Documentation

3.3.2.1 ConnectionManager()

```
ConnectionManager::ConnectionManager (
    const std::string & server_addr,
    uint16_t port )
```

Definition at line 12 of file connection_manager.cpp.

References port_, server_addr_, and socket_.

3.3.2.2 ~ConnectionManager()

```
ConnectionManager::~ConnectionManager ( ) [virtual]
```

Definition at line 49 of file connection_manager.cpp.

References socket_.

3.3.3 Member Function Documentation

3.3.3.1 authenticate()

```
void ConnectionManager::authenticate (
    const std::string & login,
    const std::string & password )
```

Definition at line 55 of file connection_manager.cpp.

References AuthManager::computeHash(), AuthManager::generateSalt(), receiveString(), and sendString().

Referenced by main().

3.3.3.2 processVectors()

```
std::vector< int32_t > ConnectionManager::processVectors (
    const std::vector< std::vector< int32_t >> & vectors )
```

Definition at line 73 of file connection_manager.cpp.

References `receiveData()`, and `sendData()`.

Referenced by `main()`.

3.3.3.3 receiveData()

```
void ConnectionManager::receiveData (
    void * data,
    size_t size ) [protected], [virtual]
```

Definition at line 111 of file connection_manager.cpp.

References `socket_`.

Referenced by `processVectors()`.

3.3.3.4 receiveString()

```
std::string ConnectionManager::receiveString (
    size_t max_length = 1024 ) [protected], [virtual]
```

Definition at line 131 of file connection_manager.cpp.

References `socket_`.

Referenced by `authenticate()`.

3.3.3.5 sendData()

```
void ConnectionManager::sendData (
    const void * data,
    size_t size ) [protected], [virtual]
```

Definition at line 98 of file connection_manager.cpp.

References `socket_`.

Referenced by `processVectors()`, and `sendString()`.

3.3.3.6 sendString()

```
void ConnectionManager::sendString (
    const std::string & str ) [protected], [virtual]
```

Definition at line 127 of file `connection_manager.cpp`.

References `sendData()`.

Referenced by `authenticate()`.

3.3.4 Member Data Documentation

3.3.4.1 port_

```
uint16_t ConnectionManager::port_ [protected]
```

Definition at line 32 of file `connection_manager.h`.

Referenced by `ConnectionManager()`.

3.3.4.2 server_addr_

```
std::string ConnectionManager::server_addr_ [protected]
```

Definition at line 31 of file `connection_manager.h`.

Referenced by `ConnectionManager()`.

3.3.4.3 socket_

```
int ConnectionManager::socket_ [protected]
```

Definition at line 30 of file `connection_manager.h`.

Referenced by `ConnectionManager()`, `receiveData()`, `receiveString()`, `sendData()`, and `~ConnectionManager()`.

The documentation for this class was generated from the following files:

- [src/connection_manager.h](#)
- [src/connection_manager.cpp](#)

3.4 FileHandler Class Reference

Статический класс для операций с файлами

```
#include <file_handler.h>
```

Static Public Member Functions

- static std::vector< std::vector< int32_t > > [readVectors](#) (const std::string &filename)
- static void [saveResults](#) (const std::string &filename, const std::vector< int32_t > &results)
- static std::pair< std::string, std::string > [readConfig](#) (const std::string &filename)

Private Member Functions

- [FileHandler](#) ()=delete
- [~FileHandler](#) ()=delete
- [FileHandler](#) (const [FileHandler](#) &)=delete
- [FileHandler](#) & [operator=](#) (const [FileHandler](#) &)=delete

3.4.1 Detailed Description

Статический класс для операций с файлами

Definition at line 18 of file file_handler.h.

3.4.2 Constructor & Destructor Documentation

3.4.2.1 FileHandler() [1/2]

```
FileHandler::FileHandler ( ) [private], [delete]
```

3.4.2.2 ~FileHandler()

```
FileHandler::~~FileHandler ( ) [private], [delete]
```

3.4.2.3 FileHandler() [2/2]

```
FileHandler::FileHandler (
    const FileHandler & ) [private], [delete]
```

3.4.3 Member Function Documentation

3.4.3.1 operator=()

```
FileHandler& FileHandler::operator= (
    const FileHandler & ) [private], [delete]
```

3.4.3.2 readConfig()

```
std::pair< std::string, std::string > FileHandler::readConfig (
    const std::string & filename ) [static]
```

Definition at line 70 of file `file_handler.cpp`.

Referenced by `main()`.

3.4.3.3 readVectors()

```
std::vector< std::vector< int32_t > > FileHandler::readVectors (
    const std::string & filename ) [static]
```

Definition at line 11 of file `file_handler.cpp`.

Referenced by `main()`.

3.4.3.4 saveResults()

```
void FileHandler::saveResults (
    const std::string & filename,
    const std::vector< int32_t > & results ) [static]
```

Definition at line 50 of file `file_handler.cpp`.

Referenced by `main()`.

The documentation for this class was generated from the following files:

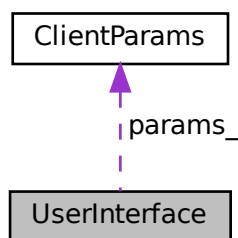
- [src/file_handler.h](#)
- [src/file_handler.cpp](#)

3.5 UserInterface Class Reference

Класс для обработки параметров командной строки

```
#include <user_interface.h>
```

Collaboration diagram for UserInterface:



Public Member Functions

- [UserInterface](#) ()
- bool [parseArguments](#) (int argc, char **argv)
- [ClientParams](#) [getParams](#) () const
- std::string [getHelp](#) () const

Private Attributes

- boost::program_options::options_description [desc_](#)
- [ClientParams](#) [params_](#)

3.5.1 Detailed Description

Класс для обработки параметров командной строки

Definition at line 26 of file user_interface.h.

3.5.2 Constructor & Destructor Documentation

3.5.2.1 UserInterface()

```
UserInterface::UserInterface ( )
```

Definition at line 10 of file user_interface.cpp.

References [ClientParams::config_file](#), [desc_](#), [ClientParams::input_file](#), [ClientParams::output_file](#), [params_](#), [ClientParams::port](#), and [ClientParams::server_address](#).

3.5.3 Member Function Documentation

3.5.3.1 getHelp()

`std::string UserInterface::getHelp ( ) const`

Definition at line 55 of file `user_interface.cpp`.

References `desc_`.

Referenced by `main()`.

3.5.3.2 getParams()

[ClientParams](#) `UserInterface::getParams ( ) const` [inline]

Definition at line 30 of file `user_interface.h`.

References `params_`.

Referenced by `main()`.

3.5.3.3 parseArguments()

```
bool UserInterface::parseArguments (
    int argc,
    char ** argv )
```

Definition at line 20 of file `user_interface.cpp`.

References `desc_`.

Referenced by `main()`.

3.5.4 Member Data Documentation

3.5.4.1 desc_

`boost::program_options::options_description UserInterface::desc_` [private]

Definition at line 34 of file `user_interface.h`.

Referenced by `getHelp()`, `parseArguments()`, and `UserInterface()`.

3.5.4.2 params_

[ClientParams](#) `UserInterface::params_` [private]

Definition at line 35 of file `user_interface.h`.

Referenced by `getParams()`, and `UserInterface()`.

The documentation for this class was generated from the following files:

- [src/user_interface.h](#)
- [src/user_interface.cpp](#)

Глава 4

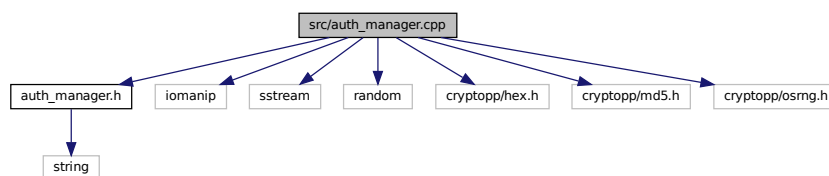
File Documentation

4.1 src/auth_manager.cpp File Reference

Реализация класса аутентификации

```
#include "auth_manager.h"  
#include <iomanip>  
#include <sstream>  
#include <random>  
#include <cryptopp/hex.h>  
#include <cryptopp/md5.h>  
#include <cryptopp/osrng.h>
```

Include dependency graph for auth_manager.cpp:



4.1.1 Detailed Description

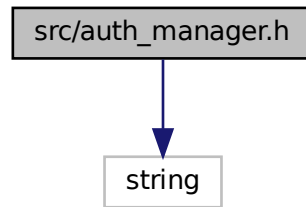
Реализация класса аутентификации

4.2 src/auth_manager.h File Reference

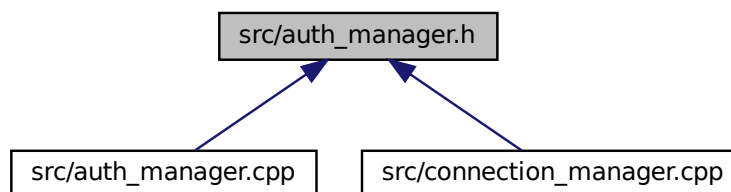
Заголовочный файл класса аутентификации

```
#include <string>
```

Include dependency graph for `auth_manager.h`:



This graph shows which files directly or indirectly include this file:



Classes

- class [AuthManager](#)
Статический класс для криптографических операций

Macros

- `#define` [CRYPTOPP_ENABLE_NAMESPACE_WEAK](#) 1

4.2.1 Detailed Description

Заголовочный файл класса аутентификации

4.2.2 Macro Definition Documentation

4.2.2.1 CRYPTOPP_ENABLE_NAMESPACE_WEAK

```
#define CRYPTOPP_ENABLE_NAMESPACE_WEAK 1
```

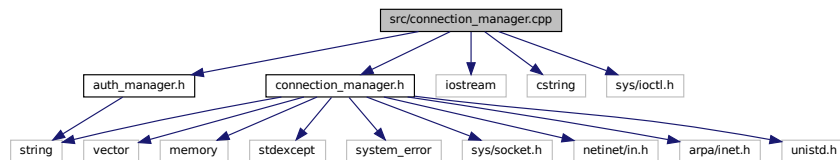
Definition at line 9 of file auth_manager.h.

4.3 src/connection_manager.cpp File Reference

Реализация класса управления соединением

```
#include "connection_manager.h"
#include "auth_manager.h"
#include <iostream>
#include <cstring>
#include <sys/ioctl.h>
```

Include dependency graph for connection_manager.cpp:



4.3.1 Detailed Description

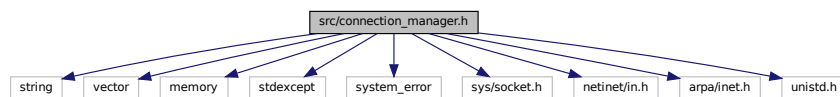
Реализация класса управления соединением

4.4 src/connection_manager.h File Reference

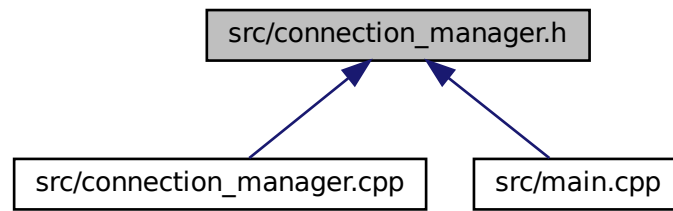
Заголовочный файл класса управления соединением

```
#include <string>
#include <vector>
#include <memory>
#include <stdexcept>
#include <system_error>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <unistd.h>
```

Include dependency graph for connection_manager.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [ConnectionManager](#)
Класс для управления TCP-соединением с сервером

4.4.1 Detailed Description

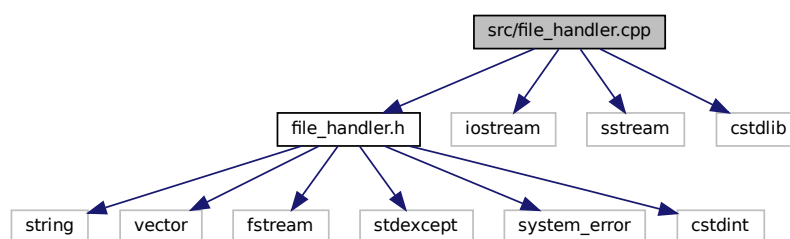
Заголовочный файл класса управления соединением

4.5 src/file_handler.cpp File Reference

Реализация класса работы с файлами

```
#include "file_handler.h"
#include <iostream>
#include <sstream>
#include <cstdlib>
```

Include dependency graph for `file_handler.cpp`:



4.5.1 Detailed Description

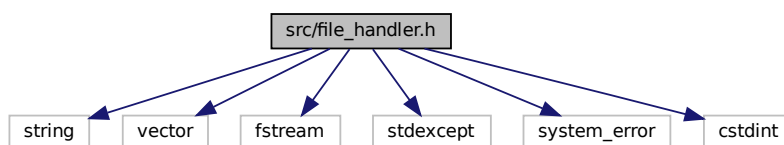
Реализация класса работы с файлами

4.6 src/file_handler.h File Reference

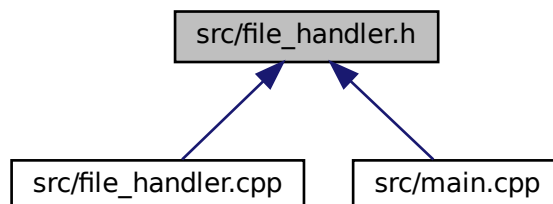
Заголовочный файл класса работы с файлами

```
#include <string>
#include <vector>
#include <fstream>
#include <stdexcept>
#include <system_error>
#include <cstdint>
```

Include dependency graph for file_handler.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [FileHandler](#)
Статический класс для операций с файлами

4.6.1 Detailed Description

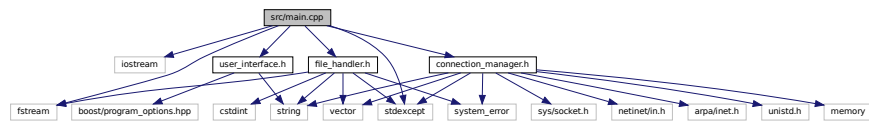
Заголовочный файл класса работы с файлами

4.7 src/main.cpp File Reference

Главный модуль клиентского приложения

```
#include <iostream>
#include <stdexcept>
#include "user_interface.h"
#include "file_handler.h"
#include "connection_manager.h"
#include <fstream>
```

Include dependency graph for main.cpp:



Functions

- int `main` (int argc, char **argv)
Точка входа в программу

4.7.1 Detailed Description

Главный модуль клиентского приложения

4.7.2 Function Documentation

4.7.2.1 main()

```
int main (
    int argc,
    char ** argv )
```

Точка входа в программу

Parameters

argc	Количество аргументов
argv	Массив аргументов

Returns

0 при успехе, 1 при ошибке

Definition at line 19 of file main.cpp.

References `ConnectionManager::authenticate()`, `UserInterface::getHelp()`, `UserInterface::getParams()`, `UserInterface::parseArguments()`, `ConnectionManager::processVectors()`, `FileHandler::readConfig()`, `FileHandler::readVectors()`, `FileHandler::saveResults()`, and `ClientParams::server_address`.

4.8 src/user_interface.cpp File Reference

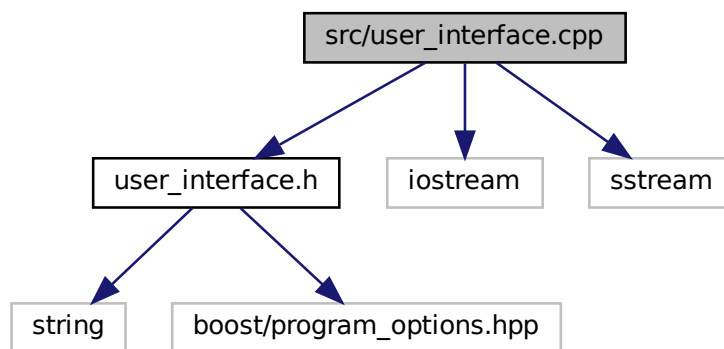
Реализация класса пользовательского интерфейса

```
#include "user_interface.h"
```

```
#include <iostream>
```

```
#include <sstream>
```

Include dependency graph for user_interface.cpp:



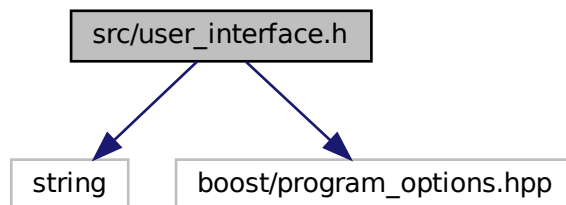
4.8.1 Detailed Description

Реализация класса пользовательского интерфейса

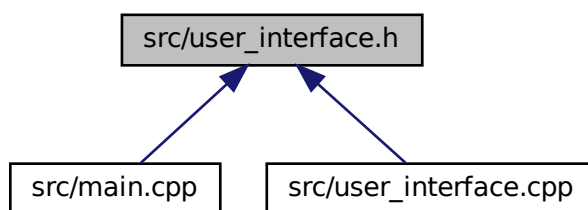
4.9 src/user_interface.h File Reference

Заголовочный файл класса пользовательского интерфейса

```
#include <string>
#include <boost/program_options.hpp>
Include dependency graph for user_interface.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- struct [ClientParams](#)
Структура для хранения параметров клиента
- class [UserInterface](#)
Класс для обработки параметров командной строки

4.9.1 Detailed Description

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